

Long division

Long division is usually used when the divisor is at least two digits long and the dividend is at least 3 digits long. Unlike short division, all the workings out are written out in full below the dividing line. Multiplication is used for finding remainders. A long division sum is presented in the example on the right.

the dividing line is used in place of $\div$ or $/$ sign

The answer (or quotient) goes in the space above the dividing line.

DIVISOR
number is used to divide dividend

DIVIDEND
number that is divided by another number

The calculations go in the space below the dividing line.

put result of second division above last digit being divided into

Begin by dividing the divisor into the first two digits of the dividend. 52 fits into 75 once, so put a 1 above the dividing line, aligning it with the last digit of the number being divided.

Work out the first remainder. The divisor 52 does not divide into 75 exactly. To work out the amount left over (the remainder), subtract 52 from 75. The result is 23.

Now, bring down the last digit of the dividend and place it next to the remainder to form 234. Next, divide 234 by 52. It goes four times, so put a 4 next to the 1.

multiply 4 (the number of times 52 goes into 234) by 52 to get 208

amount left over from second division

add a decimal point then a zero

bring down zero and join it to remainder

add decimal point above other one

put result of last sum after decimal point

Work out the second remainder.

The divisor, 52, does not divide into 234 exactly. To find the remainder, multiply 4 by 52 to make 208. Subtract 208 from 234, leaving 26.

There are no more whole numbers to

bring down, so add a decimal point after the dividend and a zero after it. Bring down the zero and join it to the remainder 26 to form 260.

Put a decimal point after the 14.

Next, divide 260 by 52, which goes five times exactly. Put a 5 above the dividing line, aligned with the new zero in the dividend.